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Scaling Circularity in Disposal Systems: Modeling Large-Scale Feasibility & Net Gains

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1 Context

1.1 The Linear Waste Processing Problem

Linear, one-directional waste processing contributes significantly to greenhouse gas (GHG) emissions, largely due to indiscriminate collection and processing streams [1]. As the effects of such emissions have become increasingly pronounced with the rising rate of climate change, various countries and regions across the globe have adopted new ways to manage waste. Broadly, the aim of such modification involves maximizing resource lifespan and minimizing harmful emissions.

1.2 Global Mitigation Efforts: Small Scale Systems that Work

Currently, many such efforts to systematically change the way waste streams are collected and processed are concentrated in various parts of Europe and Asia. For instance, German and Swiss waste systems frequently implement an early and persistent division of over five waste streams. Although there is a degree of variation in exactly how many streams are collected in each internal municipality, the following waste channels are typically distinguished: organic biowaste, paper & cardboard, glass, plastics/metal/packaging, and general waste [13][14].

While the effects of these systems depend on additional variables—including but not limited to consumer participation incentives and behavioral factors, processing efficiency of each stream, and degree of standardization across municipalities—both systems yield over 50% of municipal waste being recycled or reused, with a recent German report suggest an increase to over 67% of waste being recycled or reused [13][14]. Additionally, both systems report the proportion of total waste that ends up in landfills to be under 1% [13][14]. In addition to improved waste classification, this low landfill rate may be further explained by energy recovery through waste-to-energy (WtE) incineration. While incineration may have harmful air pollution implications if uncontrolled, the combination of upstream sorting—which prevents toxic materials from being burned—with carefully designed plants to filter and monitor incinerated materials make this an environmentally feasible option.

1.3 Waste Management in the USA

In contrast, USA reports suggest that in 2018 only around 32% of municipal waste was recycled or composted, with around 50% of total waste going to landfills [3]. These lower rates may be at least partially explained by the lack of standardized waste management protocols. The ability and willingness to enforce such policy may be influenced by the nation's larger scale, social and political influences, low incentives for consumer

participation, and funding constraints. Even existing programs observe limited efficacy; a 2020 report by The Recycling Partnership suggests that only around 32% of the estimated 37.4 million tons of single-family recyclables are successfully captured and processed [3].

In addition to inefficiencies in existing US recycling systems, poor or non-existent waste classification results in large quantities of organic material ending up in landfills, where they decompose through anaerobic digestion. Anaerobic digestion occurs when organic material breaks down without oxygen, which occurs when waste is compacted [1]. This is a large determinant in the 17% of total anthropogenic methane emissions in the US that is attributed to landfills [1].

1.4 Importance of Waste Stream Classification

In contrast, a distinguished classification stream for organic material would prevent such anaerobic digestion. Alternatively, these materials will undergo aerobic digestion, which produces negligible concentrations of methane, or controlled Anaerobic Digestion (AD), under which over 99% of released methane can be captured and converted to biogas energy for thermal and fuel needs [12]. Distinctions within the organics determine which alternative is more productive, although the latter may be more costly.

The upstream classification of recycled materials plays an important role in achieving efficient and significant results, as cross-stream contamination at any point of along waste material streams makes it difficult—if not impossible—to sort. Even contamination between recyclable materials like glass and paper often causes significant material degradation, decreasing the quality of the reusable output [5]. Thus, it is equally important that those streams be properly managed and separated beyond the household classification level. An effective waste management system must consider the following three layers, at a minimum: per-household management, transportation management, and processing management.

1.5 Scaling Solutions: The Role of Big Data

Meaningfully evaluating the feasibility and effectiveness of such a system in a large and heterogeneous nation like the USA involves complex analyses of a vast range of data. While the success of a rigorous waste classification system implies meaningful environmental benefits, the system does not come free of costs. In addition to evident monetary costs associated with new infrastructure and planning efforts, poor transportation logistics and processing inefficiencies may offset even the environmental gains through alternative channels of pollutants and land use.

Assessing the feasibility and effectiveness of a large-scale, optimized waste management system requires analyzing a wide range of interconnected data across multiple layers of the waste stream. Analysis occurs across the consumer, transport, and processing levels, aggregating and relating datasets such as waste processing facility efficiency, transportation networks, municipal waste generation, material market demand, measured environmental impacts of existing waste management, and demographic indicators. Applying big data techniques allows cross-sector analysis to better understand, quantify, and project per-layer environmental and economic impacts of implementing an optimized waste management system.

Such analysis highlights the relevance of big data: the scale, variety, and data interconnectivity of the waste management analysis demand sophisticated modeling and computational resources before actionable conclusions can be drawn.

1.6 Investment in a Better Future

Waste management targets only a small portion of climate-related emissions. Changes in the waste sector are often considered relatively inexpensive or easy to implement compared to environmentally motivated changes across other economic sectors. However, a comprehensive quantification of the associated impact may provide a tangible point of comparison with investments in energy or agriculture, two leading emission sectors. Investing in a deeper understanding of these costs and benefits offers potential to avoid major long-term financial losses while supporting effective, evidence-based strategies for climate mitigation across environmental, economic, and social spheres.

Large-scale analysis facilitates a comprehensive understanding of these holistic costs and benefits, which appeals not only to the environmentalist, but also more broadly to economists, urban planners, and policy makers seeking to optimize spending to maximize benefits. Furthermore, as emerging technologies—such as AI-driven household waste sorting—become increasingly feasible and affordable, it is essential to understand the broader system that such innovations contribute to or depend on. Thus, this analysis is increasingly relevant to technology developers and stakeholders who wish to optimize investment in green technology.

1.7 Personal Motivation and Qualifications

My interest in this topic is strongly motivated by a desire to explore areas in which data techniques and emerging technologies can be applied to global issues such as climate change mitigation. In particular, waste management presents an interesting and complex problem well-suited for data-driven analysis, as it involves a broad spectrum of participants within a vastly interconnected, circular system.

My background in both quantitative economics and computer science informs my understanding of interdisciplinary data modeling, cost minimization, and big data techniques. By facilitating a deeper understanding of system-level tradeoffs and quantifying the impacts of key variables, I aim to support effective and thoughtful pathways for technological innovation that balance both short- and long-term impacts.

2 Introduction

2.1 Existing Solutions

In order to analyze projected impacts of a more efficient waste management system, we have developed a framework that is largely informed by success in current German and Swiss waste management systems. The selection of such systems comes from their associated environmental impacts, the availability of relevant data, and a conceptual similarity to the proposed US framework. The largest difference to the proposed framework—and a distinguishing determinant of the proposed framework’s feasibility and projected impacts—is that these models were applied in smaller, more homogenous environments, which likely facilitated administrative ability to enforce and incentivize consumer participation, simplified transportation route logistics and proximity concerns, and reduced the magnitude of associated financial costs.

While such differences may reasonably inform a belief that such a system is infeasible in a country like the US, the increasingly viable development of household-level, smart sorting reduces this burden significantly, particularly with respect to behavioral incentives and difficulties enforcing widespread participation. Presently, cameras, weight sensors, and visual recognition systems have been developed, enabling proper disposal recommendations based on automated detection analysis [18][19]. Although these early commercial models and research prototypes still have limited usability and largely rely on user compliance with the given sorting guidance, ongoing technological and AI developments instill optimism that a reliable, fully automated sorting bin—removing consumer responsibility almost completely—may be reasonably achievable.

Thus, to simplify the proposed model for the sake of more objective quantification, the proposed framework assumes perfect consumer participation. Thus, both the design of the base framework and the input optimization model focus on logistical, infrastructural, financial, and labor demands in the transportation and processing layers rather than more subjective household layer measures such as consumer incentives, political ideology, and other demographic influences that might affect individual willingness to participate beyond measurable feasibility.

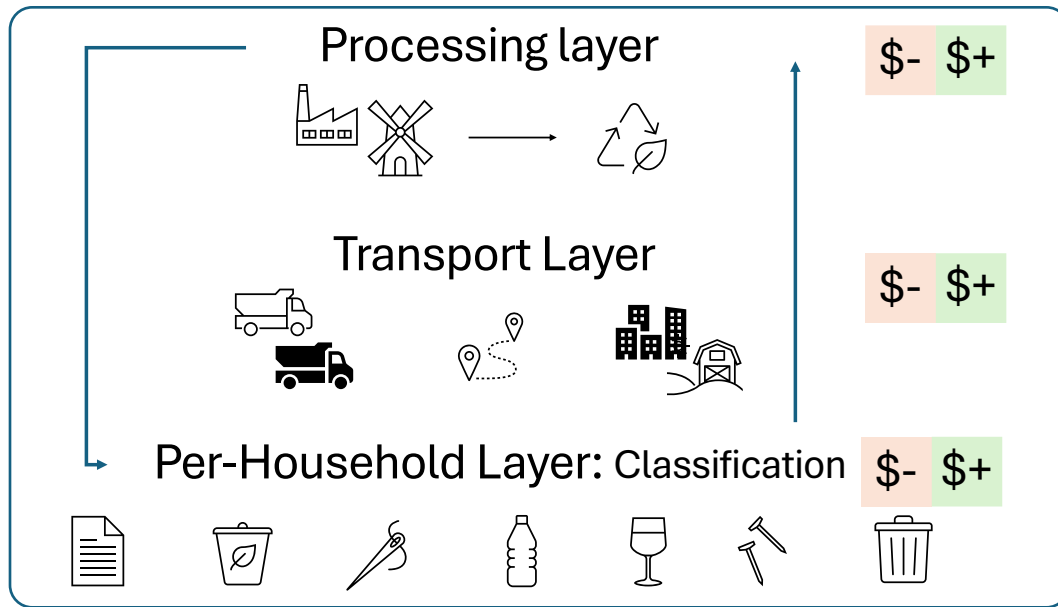


Figure 1. *Waste-Stream System Layer Diagram*

2.2 Innovation Proposal

Building on these existing systems and emerging technologies, the proposed US system introduces a waste management framework with more efficient and distinct waste classification streams that promotes material circularity and makes recommendations to maximize environmental, economic, and social outcomes.

Like the German and Swiss frameworks discussed, the proposed base framework involves a separation of organic waste, textiles, paper, plastics, metals, and glass. However, this base framework is then algorithmically adjusted according to regressive analysis across a broad range of big data inputs, optimizing processing pathways for each stream and breaking down the results in a digestible way. Thus, this approach differs from existing systems by integrating dynamic, multi-layer optimization techniques that use large-scale data to refine parameters based on a unique awareness of key variables across different disciplines.

The model's overall flow and details of proposed big data analysis techniques are further discussed in later sections.

2.3 Practical Implementation

The implementation of this solution requires continuous access to diverse, up-to-date data sets both for initial development of the optimization algorithm and for continued accuracy as economic, technological, and social conditions evolve over time. The composition of the necessary data sets is detailed in later sections. Moreover, because the model's

predictive analysis relies on historical pattern detection and variable analysis over time, this data must be durable. Thus, this will likely require large-scale storage solutions, which may invoke additional costs.

Expert advice from economists, urban planners, and environmentalists may be valuable to inform the base framework design in a practical implementation. Moreover, development of the optimization evaluation model requires technical developers with expertise in data science and software development.

These resource requirements have been identified to fit the project's goals of accurate and comprehensive cost prediction, future use scalability, and technological feasibility.

2.4 Benefits to Humankind

The waste sector in countries like the US contributes significantly to harmful greenhouse gas emissions, particularly with respect to methane emissions. This proposed solution aims to facilitate a better understanding of how different systematic components can magnify preventable environmental and financial strain. By quantifying the various dimensions of waste management system efficiency, this project proposes a potential for a more pragmatic and evidence-based approach to system design that can have global short- and long-term benefits. These benefits may include positive impacts on air quality and public health through reduced methane and particulate emissions, and natural resource conservation through a decreased reliance on virgin material extraction. Additionally, the model may contribute to positive downstream economic effects of reduced material costs, potential for longer-term reduction in environmental reparation costs, and the generation of new technical and operational jobs to support system planning and collection demands.

Moreover, while this analysis focuses on scaling solutions within the US, the model is developed generally and flexibly to facilitate future use for any nation given proper parameter adjustments.

3 Technical Design

3.1 Relevant Datasets

An extensive list of relevant data categories that should be accessible to the model is organized below. This was determined through careful research and analysis of potential influences—direct and indirect—that may affect the success of a waste management system. Nevertheless, this is an initial proposal, and a final implementation of the base model may be further extended through expert consultation in the following sectors.

We present the following relevant data categories that have been evaluated for relevance and availability. These have been organized for interpretability but note that some data sets may have implications across identified categories.

Infrastructure and Efficiency

- ❖ Locations of existing waste processing facilities in the US and their associated capacity, efficiency, and % quality degradation of recycled materials [3][4]
- ❖ Location distribution and efficiency data for alternative processing facility types and implementations, including % quality degradation of recycled materials [3]
- ❖ Costs of building current waste processing and transport infrastructure [22]
- ❖ Data describing infrastructure social and health impacts on neighboring communities [23]
- ❖ Emission and energy costs for each type of processing facility [1][17]

Transportation Logistics

- ❖ Residential location distribution in the USA [10]
- ❖ Existing USA transportation networks [9]

Labor Demand

- ❖ Waste management labor force in the USA [21]
- ❖ Current unemployment statistics in the USA [21]
- ❖ Economic effects of jobs in waste management sector [23]
- ❖ Recorded municipal waste generation across USA municipalities [3]

Reused Material Market Demand

- ❖ Reused material costs and market demand [6]
- ❖ Virgin material market demand [7][11]
- ❖ Virgin material extraction and production costs [11]

Environmental Impact Quantification

- ❖ Swiss waste quantified emission reductions [16]
- ❖ German waste quantified emission reductions [13]
- ❖ Measured emission and air quality impacts of current US waste management system [1][8]
- ❖ Air quality degradation per transport unit, waste transport vehicle efficiency details [17]

3.2 Big Data Techniques:

3.2.1 Storage

Our input data requires a large-scale storage solution for relevant and persistent data such as geospatial household coordinates, transportation graphs, facility operation and efficiency data, emissions statistics, and material market trends data.

This requires support for parallel reads—to access data for optimization analysis—and writes—to update or add to existing data sets. Thus, we propose the use of *Hadoop Distributed File System (HDFS)*; the system will run on a distributed compute-cluster with N worker nodes, each storing partitions of HDFS blocks and Spark RDDs. This architecture enables horizontal scaling, fault tolerance, and high-throughput processing during optimization.

HDFS is selected over alternatives such as GFS primarily due to its native integration with Spark and its suitability for open-source distributed computation environments.

3.2.2 Compute

The proposed model relies on the ability to perform large-scale iterative optimizations and transformations across large and diverse datasets, which will ultimately converge toward a minimized total cost function that considers environmental, financial, and economic dimensions.

The use of *Spark* will allow in-memory distributed execution to optimize processing facility locations (geo-coordinates) and waste stream transport routes for household clusters, simulating different input combinations and running graph analyses on transportation networks while calculating large-scale cost relationships in parallel. This large-scale parallel computation is achievable through Spark's *Resilient Distributed Datasets (RDDs)* [24]. Thus, parameter adjustments enable iterative refinements to yield an optimized infrastructure and logistical framework. This goal is supported through Spark's compatibility with the following iterative algorithms.

3.2.3 Algorithms

LocationRank

The US transportation network can be represented as graphs, which may be processed through a modified *PageRank* algorithm [20] that can identify heavy traffic hotspots to yield optimal paths considering system-wide travel costs. We will refer to this modified algorithm as *LocationRank*, as this will ultimately "rank" optimal processing facilities.

The incoming and outgoing "links" of a processing facility will be the routes themselves, and these will be evaluated based on travel distance, traffic burdens, emission costs, etc. Thus, the model will effectively use output rankings to prioritize candidate facility locations.

K-Center Clusters

Moreover, rather than treating each household as an individual system node, our model will use *K-center clustering* to optimally partition households into zones that will need to be serviced for each waste stream collection [25].

Household assignment to a zone will be based on minimizing household distance to its nearest "center" point, where centers correspond to the *K* US-processing facility locations that have been algorithmically determined.

Furthermore, the use of *Bloom filters* may be valuable to reducing lookup latencies in computations that involve frequently accessing household-to-cluster relationships.

3.2.4 Result Breakdown and Front-Facing Output

Once the model's parameter optimization has run, these results must be transformed into an interpretable dashboard that users can use to better understand cost relationships.

We propose the use of *BigQuery* to support low-latency SQL querying over optimization results, which can be used to populate dashboard sections through rule-based report-generation templates. Thus, this will allow users to better understand the relevance and implications of each optimized parameter.

3.4 End Product and Intended Usage

The final product will be an interactive dashboard that quantifies total net costs, net environmental costs, net economic costs, and monetary investment requirements. The dashboard will break down optimized parameter values, offering different recommendations within each input category and explanations of that parameter's contribution to different cost categories. These recommendations may include insight on specifications of a processing facility to meet certain efficiency standards, optimal vehicle and fuel types, etc.

By default, the system will optimize for total net costs. However, users may adjust input quantities and observe real-time changes across different dimensions. Moreover, users may set fixed base costs (for example, to represent a base cost of household-level sorting bin automation) and adjust weights of cost categories to fit their priorities. For instance, an environmentalist who may be more willing to absorb monetary costs may choose to solely

optimize environmental costs, while a financial advisor or an economist may choose to adjust the weights differently. Users may reset to the default optimization at any time.

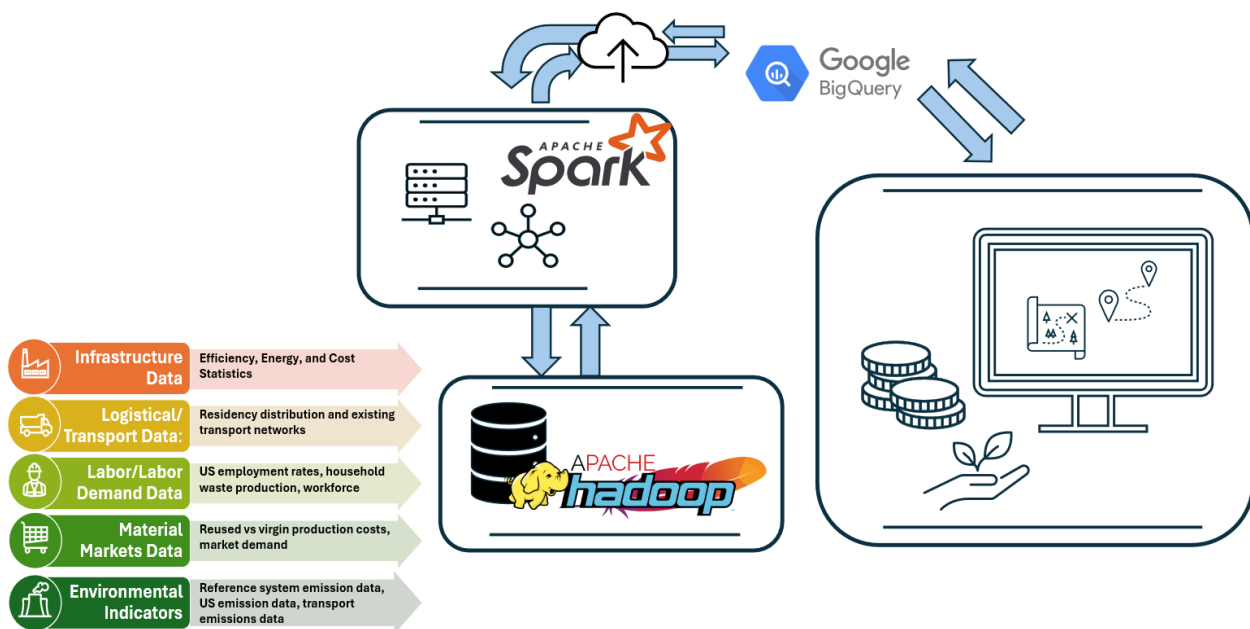


Figure 2. *Simplified System Architecture*

4 Work Plan

4.1 Implementation Phases

Phase 1: Initialize Distributed Database

Goal: Aggregate diverse data into a distributed storage system that can be efficiently accessed by the model.

Plan:

- Collect relevant data sets and data sets and standardize CSV format or graph structures (e.g., transportation networks transformed via *Spark GraphX*).
- Store all datasets in HDFS and deploy storage layer on the distributed cluster.

Evaluation: Structured, queryable data sets are successfully added to HDFS and can be accessed through parallel read operations.

Phase 2: Initialize Processing Infrastructure

Goal: Set up compute framework to support distributed and iterative algorithmic computations.

Plan: Configure and deploy an N-node Spark cluster. Optionally, monitor and manage cluster with Kubernetes.

Evaluation: Test that parallel read/write requests successfully access and modify HDFS.

Phase 3.1: Implement Cost Functions

Goal: Develop cost functions that the model will use to minimize loss and maximize benefits.

Plan: Develop a mathematical net-cost function that encompasses the relationship of direct investment, quantified environmental impact, quantified economic impact, and long-term projects given a discount rate.

Evaluation: Test cost function in controlled setting with known data.

Phase 3.2: Implement Optimization Algorithms

Goal: Develop LocationRank and K-Center clustering algorithms that are compatible with Spark.

Plan: Adapt PageRank algorithms and leverage existing libraries to develop algorithms that adequately weigh relevant variables in the project's context.

1. LocationRank
 - a. Construct transportation graph.
 - b. Implement weighted PageRank in Spark using RDDs.
 - c. Compute candidate facility location score by considering relevant transportation factors (see specifications in *Technical Solution 3.2.3*).
2. K-Center Clustering
 - a. Implement K-center clustering for national household groupings.
 - b. Parallelize with Spark RDDs.
 - c. Integrate Bloom filters to store household-to-cluster membership for fast lookup.

Evaluation: Test algorithms in controlled setting with known graph data to ensure accurate rankings and cluster assignments.

Phase 4: Implement Distributed Optimization Model

Goal: Configure a model that consistently optimizes cost

Plan:

1. Build an optimization loop that iteratively adjusts:
 - a. Number of processing streams (defaults to hold seven proposed streams constant)
 - b. Number of workers for each type of worker
 - c. Number of processing facilities for each stream
 - d. Optimal location coordinates for each processing facility
 - e. % trash directed to landfill
 - f. % trash incinerated and converted to energy
 - g. Pickup frequency for each stream
 - h. Number of household clusters
2. Implement convergence logic through monitoring cost improvements given a defined threshold.

Evaluation: Test distributed optimization engine on smaller and controlled data sets to ensure that convergence is consistently and accurately achieved.

Phase 5: Integrate Components

Goal: Fully integrate backend system components.

Plan: Connect optimization engine, algorithms, cost functions, and data sources into a single distributed pipeline.

Evaluation: Run optimization tests, iteratively increasing data and cluster sizes to ensure scalability, consistent behavior, and fault tolerance.

Phase 6: Configure BigQuery

Goal: Enable analytical queries over exported optimization results.

Plan:

- Export optimization outputs from HDFS into BigQuery.
- Define BigQuery schemas for optimized values, costs, cluster assignments, processing facility scores, etc.

Evaluation: Run controlled test queries to ensure that each schema successfully accesses up-to-date optimization results.

Phase 7: Develop Dashboard Interface

Goal: Display results in interactive, digestible way that supports deeper understanding of how each optimized parameter plays into cost analyses.

Plan:

- Build dashboard UI using tools such as Tableau.
- Include responsive parameter adjustment elements, responsive reset and rerun buttons, and map visualizations for waste stream routing and processing facility locations.
- Connect dashboard with BigQuery.

Evaluation: Run model and verify that dashboard elements are responsive, well-designed. Run tests to ensure that information is consistent with BigQuery.

4.2 Measure of Success

The model ultimately aims to better quantify downstream and long-term gains to more holistically analyze whether a circular, optimized waste system is feasible at a national scale. Success is determined by the model's ability to quantify net gains and costs across different system components to identify key variables on a net outcome. The magnitude of potential gains may valuably inform various related economic and environmental assessments to assess tradeoffs of investing equivalent resources in other sectors. Results can guide policy, infrastructure planning, and economic-environmental assessments for more effective U.S. waste management.

Success is also defined by the model's usefulness to stakeholders—policymakers, urban planners, and environmental analysts—who require transparent cost breakdowns to inform infrastructure and policy decisions. Although this proposal aims to design a more efficient waste system that can offer positive short and long-term benefits, it is important to note that the optimization results may ultimately fail to support the adoption of this

system. That is, while it is entirely possible—and not unlikely—for a successful system to determine that the gains of the proposed waste framework justify the effort needed to practically implement such a system, this is not a requirement for success. Rather, the system aims to provide a deeper, evidence-based understanding of how different variables influence comprehensive costs so that optimal solutions can be identified. An optimal solution may require adjustments to the framework—perhaps the number of streams, scope of the system, or other changes—or the development of a different solution altogether, which may be better compared to or informed by the model’s analyses.

Thus, the success of the system may be better assessed through performance and accuracy metrics. The former may involve assessment of scalability by measuring runtime across cluster sizes, convergence stability and consistency in optimization recommendations, and tolerance to missing or noisy data. Accuracy assessment may further involve testing the model on past US data and existing waste systems frameworks to compare predicted results with actual realized costs. Other tests may involve running the model on past German or Swiss data and assessing whether the predictions are significantly accurate at a smaller scale. Determining the model’s ability to scale may further be useful for future extensibility to apply the model assessment to other countries and environments.

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